

SLEEP 2026

BALTIMORE, MD | JUNE 14-17

A JOINT MEETING

AASM American Academy of
SLEEP MEDICINE™

 Sleep Research Society®
Advancing Sleep & Circadian Science

Longitudinal Modeling of Sleep Timing, Duration, and Social Jetlag Across the Adult Lifespan Using High-Resolution Wearable Data

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Conflict of Interest Disclosures for Speakers

To review this speaker's disclosure
information, please visit
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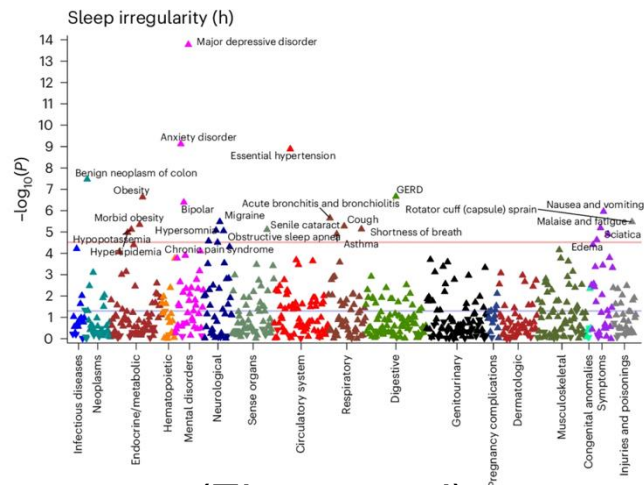
Wearables and longitudinal monitoring of sleep and circadian traits

Sleep is important to health

Most research was done using self-report or short-term monitoring

Availability of data and technology

Add Sleep PheWAS results (from published Nat Med paper)



(Zheng et al)

Wearables and longitudinal monitoring of sleep and circadian traits

Important sources of variability are well established, but never investigated within the same individual:

Extrinsic

- Week vs. Weekdays
- Seasons
- Environmental exposures

Intrinsic

- Sex, Age
- Employment Status

Study aim

Characterize demographic and environmental contributors to longitudinal variability in sleep patterns using the All of Us Research Program

All of Us Research Program

Describe program and data

Describe Fitbit measurements

Participants characteristics

Add flow chart (all AoU → included participants)

N = 42,290

Characteristic	Value
Age (years)	52 ± 17 (range 14–114)
Female	67.4%
Male	32.3%
Nights Recorded	530 ± 730 (range 1–4,171)
Average Nightly Sleep Duration (min)	447 ± 65
Average Nightly Sleep Duration (hours)	7.5 ± 1.1

Sleep and Circadian Traits x Age, Sex

Sleep and Circadian Traits x Other Sociodemographics

Sleep and Circadian Traits x Environmental Variables

Conclusions and Future Directions

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